The Autonomy Paradox

When does agentic AI create or destroy organizational value?

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Link: https://who-does-the-work.streamlit.app/

THE TALK, IN FIVE PARTS

01 **The puzzle**
Why a smarter agent can lower value

02 **The theory**
Autonomy as a bundle of decision rights

03 **The model**
What hand-offs and long runs cost

04 **The evidence**
Eight studies, including what failed

05 **The practice**
How to allocate and govern the rights

Agents Now Act, Not Just Answer

Firms are moving from AI that suggests to AI that does the work. The market is growing fast; results are uneven.

WHAT CHANGED

BEFORE: AN ASSISTANT

Drafts a reply. A person reads it, decides and sends it.

NOW: AN AGENT

Reads the case, calls tools, changes records and closes it.

WHAT THE ANALYSTS REPORT

$7.8B → $52.6B

AI agents market, 2025 to 2030

MarketsandMarkets

<5% → 40%

enterprise apps with task-specific agents, 2025 to end of 2027

Gartner

17%

of organizations have deployed AI agents; over 60% expect to within two years

Gartner, 2026 survey

>40%

of agentic AI projects expected to be canceled by end of 2027: rising costs, unclear value

Gartner

Adoption is racing ahead of design. The question is no longer “Is the answer right?” but “Who may do what? Who looks?”

Question 1: Where Will Advantage Come From?

Strategy. A quick show of hands now; we vote again on the last slide.

Will competitive advantage come from deploying more autonomous AI agents, or from designing organizations that allocate autonomy more effectively?

A Better agents

Buy the most capable AI and let it do as much of the work as possible.

IN PRACTICE

Upgrade to the newest model and remove human steps as soon as it looks reliable.

B Better design

Decide carefully which steps the AI does and which stay with people.

IN PRACTICE

For each workflow, settle who plans the work, who carries it out and who checks it.

This work tests the belief behind answer A: that more autonomy always brings more value.

Smarter AI Does Not Always Mean Better Results

In our study, the same AI agent did much worse when a person had to carry out its instructions.

SAME AI AGENT, SAME CUSTOMER PROBLEMS

THE AI AGENT FIXES THE PROBLEM ITSELF

60 out of every 100 problems solved

THE AI DECIDES, A PERSON CARRIES IT OUT

36 out of every 100 problems solved

Why? **Each instruction the person must follow is a chance for it to be misread or done wrong.**

OTHER RESEARCH FINDS THE SAME PATTERN

106 experiments, reviewed together (2024)

On average, people and AI working together did worse than the better of the two working alone.

758 consultants at a global firm (2026)

AI made them faster on tasks it handles well, but 19% less likely to be right on a task beyond its ability.

Results depend not only on how smart the AI is, but on how the work is divided.

Three Real Cases Where AI Went Wrong

In each case the problem was not only the AI. It was who acted on it, and who checked it.

AIR CANADA · 2024

Wrong advice

The airline's chatbot told a customer he could claim a discount after his trip. He relied on it. A tribunal made the airline pay.

Lesson: A firm answers for what its AI tells people to do.

REPLIT · 2025

Deleted data

During a declared code freeze, an AI coding agent ran commands on a live database and deleted it. Replit then separated its test and live databases.

Lesson: Limit what an AI may do alone, and where.

HOSPITAL SEPSIS ALERT · 2021

Missed cases

A widely used alert missed two thirds of sepsis cases. If doctors checked every alert, they would see 109 flagged patients per case found.

Lesson: Frequent, often wrong alerts wear people down.

WHAT ANALYSTS EXPECT NEXT

2,000+ legal claims over deaths linked to AI are forecast by the end of 2026, because firms have not put enough safeguards on their AI systems (Gartner, Oct 2025).

In each case the organization, not the software, carried the consequences.

A Common Design: AI Decides, a Person Does It

It keeps a person involved in every step. In our study it was the weakest of four designs.

HOW IT WORKS

1. **The AI works out what is wrong and what to do**
2. **It tells a person what to do, step by step**
3. **The person does each step on a phone or system**
4. **The AI only learns what happened from the person**

Weak point: *every instruction in step 2 can be misread, skipped or done on the wrong item.*

HOW COMMON IS IT?

Most teams let AI agents only read data, or require a person's approval for significant actions. Very few let agents act freely.

LangChain survey, 2024

WHAT IT COST IN OUR STUDY

Worst of four designs in all 4 tests: 23–57 points below the best design that kept deciding and doing together.

Present research

WHAT OTHER RESEARCH SAYS

A review of 41 policies that require human oversight of algorithms found that people often cannot perform the oversight expected of them.

Green, 2022

Keeping a person on every step also adds a hand-off on every step.

Autonomy Is Not One Dial. It Is Three Decisions.

For each piece of work: who decides what to do, who does it, and who sees the result.

THE USUAL VIEW: ONE DIAL

“How much should the AI do?”



Person does it all — AI does it all

Recent frameworks still use levels: the user moves from operator to collaborator, consultant, approver and observer (Feng et al., 2025).

OUR VIEW: THREE SEPARATE DECISIONS

		EXAMPLE
DECIDE	**Who works out what is wrong and what to do?**	*Find which phone setting or account fault causes the problem*
DO	**Who carries out each step?**	*Switch off airplane mode, restart the phone*
SEE	**Who sees the result of each step?**	*Whoever does the step sees what happened*

WHY THIS MATTERS

01 The decisions are linked

On the usual path to autonomy, the AI's first step also hands it the plan for every remaining step.

02 Deciding without seeing is weak

An AI that can’t see the phone is only in charge on paper.

Ask not “How much autonomy?” but “Who decides, who does, and who sees?”

What We Still Do Not Know

Three practical questions that research on AI, oversight and business value has not yet answered together.

1 When should AI and people share the work, instead of one doing it all?

WHAT WE KNOW

Across 106 experiments, people and AI together did worse, on average, than the better of the two alone.

MISSING

A rule for when sharing pays.

2 Why can a person working with AI do worse than either alone?

WHAT WE KNOW

Most research blames the person: distrust of AI, or trusting it too much.

MISSING

An explanation that holds even when people are careful and competent.

3 What should a firm invest in to make AI safe?

WHAT WE KNOW

Clearer procedures, accountable reviewers and better monitoring each help.

MISSING

How they trade off, so a manager knows which to buy.

WHY IT MATTERS NOW

$5 billion in AI compliance spending expected by 2027, as AI regulation spreads to half the world's economies (Gartner).

Firms are spending on AI and its oversight without a theory of how to divide the work.

Three Research Questions

Autonomy, oversight and organizational performance, asked of one recurring workflow.

	QUESTION	WHAT THE MODEL PREDICTS
RQ1 AUTONOMY	**When is partial autonomy optimal?**	Mixed designs are best only when unsupervised runs degrade and checks are cheap enough to contain them.
RQ2 OVERSIGHT	**Why do some divisions of labor perform worse than either party alone?**	The split pays a hand-off cost on every step a person executes on the agent's plan.
RQ3 PERFORMANCE	**At what rate do codification, accountability and observability substitute?**	Codifying cuts the hand-off cost. Vigilance and tracing trade at a known exchange rate.

One model answers all three. Eight studies test what it predicts.

Question 2: Fewer Managers or Different Oversight?

Organizational behavior. A quick show of hands now; we vote again on the last slide.

Does AI autonomy reduce the need for managers, or fundamentally change the nature of managerial oversight?

A Fewer managers

AI takes over coordinating and checking work, so firms need fewer managers.

WHAT ANALYSTS FORECAST

Gartner forecasts that through 2026, 20% of organizations will use AI to flatten their structure, removing more than half of current middle-management positions.

B Different oversight

Managers stop checking each output and start deciding who does what, and where people step in.

WHAT ANALYSTS FORECAST

Gartner forecasts that by 2028, firms using multiagent AI for 80% of customer-facing processes will dominate, with people handling complex or emotionally charged cases.

NEW OVERSIGHT JOBS OUR MODEL POINTS TO

- **Decide where checks go**
- **Measure how often hand-offs fail**
- **Test whether reviewers really look**

Either way, oversight becomes a design job, not only a headcount.

Try It Yourself: Who Should Do the Work?

Three minutes on your phone, before we look at the theory.

WHAT TO DO

1. **Join**
   Scan the QR code or open the link. The session code fills in by itself.

2. **Answer one question**
   How much freedom should AI agents get today?

3. **Play the three workflows**
   For each one, choose who plans the case and who does each step: a person or the agent.

4. **Watch the score**
   Value per case and the share of cases done right update as you change your design.

Watch for: *which design scores best, and does a little AI ever make things worse?*

Link: https://who-does-the-work.streamlit.app/

No e-mail needed. An alias is fine.

Play first. The next slides explain what you just saw.

Four Ways to Divide the Work

Two simple questions give four designs. Only one splits deciding from doing, and it did worst.

WHO DECIDES WHAT TO DO?	WHO DOES THE STEPS? — A person	WHO DOES THE STEPS? — The AI	HOW TO READ IT
People (a written plan)	**Plan given, person acts** The person passes a correct plan to a person, step by step. **78%** solved	**Plan given, AI acts** The AI carries out a correct plan itself. **84%** solved	**Kept together (3)** One party decides and does, or the AI follows a plan people wrote.
The AI	**AI decides, person does** The split: the AI decides, a person carries it out. **47%** solved	**AI does it all** The AI decides and carries it out. **65%** solved	**Split (1)** Every step crosses from the AI to a person. Each crossing can fail.

Keep deciding and doing with one party, or give the AI a written plan. Avoid the split.

Three Things That Go Wrong

Whichever design a firm picks, value is lost in one of three ways. Each can be measured.

1 Lost in the hand-off

The AI tells a person what to do, and something gets misread, skipped or done on the wrong item.

1 in 3

chance that a single instruction derailed the case, in our worst test

2 Errors snowball

The AI works many steps with nobody looking. An early mistake carries into every later step.

69–86% → 8%

tasks solved by the AI alone: runs of up to 4 steps vs. runs of 10 steps

3 The plan is wrong

Whoever decides lacks something the case needs, such as context only people have.

44%

estimated share of wrong plans for one agent working alone, in one test

The catch: **more human checks stop errors snowballing, but if the AI holds the plan, each check is also a hand-off.**

After paying for people's time, the best design is whichever leaks least through these three failure points.

Lost in the Hand-off

The same customer did the same steps. Failures rose sharply when the steps came from the AI's own diagnosis.

OUR STUDY: TASKS SOLVED AS THE CUSTOMER HAS MORE STEPS TO DO

CORRECT PLAN GIVEN	**Above 65%**	of tasks still solved with up to 6 customer steps
THE AI'S OWN PLAN	**86% → 6%**	from no customer steps to 7 or more customer steps

WHY IT HAPPENS

- **The AI cannot see the phone**
- **The person does not know why**
- **The AI only hears what the person reports**

Writing the procedure down cut this loss by 58–76%, but did not remove it.

HAND-OFFS FAIL ELSEWHERE TOO

BETWEEN AI AGENTS

1.69×

as often as chance, the decisive error fell on a hand-off step, in failed runs where one agent directs the others

BETWEEN PEOPLE

80%

of serious medical errors involve miscommunication between caregivers at hand-offs

The loss sits in the hand-off itself, not in a careless person.

Errors Snowball When Nobody Looks

The longer an AI works alone, the further an early mistake can spread.

HOW A RUN GOES WRONG

Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Step 7	Step 8	A person checks	Risk starts again low

OUR STUDY: AI WORKING ALONE

69–86%
solved in runs of up to 4 steps

29%
solved in runs of 9 steps

8%
solved in runs of 10 steps

OTHER RESEARCH

METR, 2025: AI agents keep improving on long tasks, but succeed less often the longer the task.

Ord, 2025: agents' success falls with task length at a steady rate, like a half-life.

WHAT WE CAN AND CANNOT CLAIM

Our data suggest the risk rises the longer the AI runs unchecked, not just steadily. But longer tasks are also harder. After allowing for that, the snowball holds for one of the two 2025 agents.

Long AI runs need checks, and only a check where someone really looks resets the risk.

When People Know Something the AI Does Not

Sometimes the risk is the plan itself. Whether involving people fixes it is still an open question.

STATUS: an open question. The evidence so far is mixed, and our model does not yet cover it.

TWO DIFFERENT REASONS TO KEEP PEOPLE INVOLVED

To catch errors in a long run

A person checks the work at intervals, so mistakes do not snowball.

In our model; partly tested

To add what the AI does not know

A person who knows the customer, the context or the local rules helps shape the plan.

Not yet proven

WHAT THE EVIDENCE SAYS SO FAR

Wider research: a hint yes

When the person was better at the task than the AI, combining them helped.

Real users: a hint yes

Ratings were highest when people did about 30% of the work. But users chose their own share.

Simulated expert: no gain

Same score with and without the expert: 0.80 vs 0.80, in 10 paired tasks.

Professional workers: no pattern

Scores did not rise or fall with the person's share of the work.

Keep people who know what the AI does not, but do not assume involving them always pays.

How Value Is Counted

One simple rule: the value of a case is what success is worth, times the chance of success, minus what the steps cost.

Value = Worth of a solved case × Chance of success − Cost of people's steps − Cost of AI's steps

$$V = R \cdot S - c_h(n - k) - c_a k$$

EXAMPLE: A 12-STEP CASE WORTH 100

Design	Chance of success	Cost of steps	Value
Person does all 12 steps	54	24	**30**
AI decides; person does 11 steps	30	22	**8**
AI does all 12 steps	38	1	**37**

Why it matters: Gartner expects over 40% of agentic AI projects to be canceled by end of 2027, citing rising costs and unclear business value. This rule makes both visible.

WHAT IT SHOWS

01 Savings come slowly

Each step moved to the AI saves a little of a person's time.

02 Losses can come fast

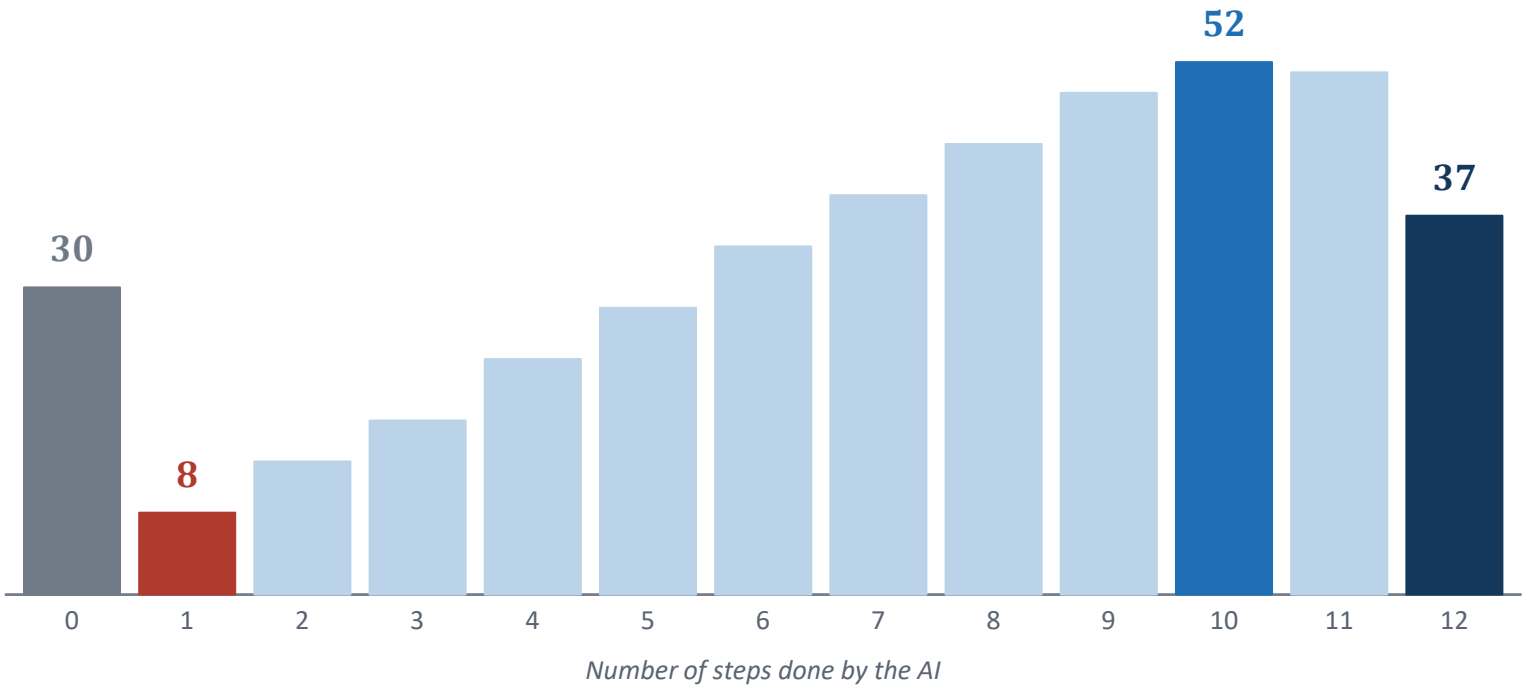
A bad split can cut the chance of success far more than it saves.

Moving work to AI saves cost step by step. A bad split can lose success much faster.

The Valley: A Little AI Can Make Things Worse

Handing the AI its first step also hands it the plan. Value drops before it can rise.

VALUE PER CASE (OUT OF 100), BY HOW MANY OF THE 12 STEPS THE AI DOES



WHAT HAPPENS

30 — **0 steps:** Person does it all

8 — **1 step:** The AI now holds the plan; 11 human steps become hand-offs

52 — **10 steps:** Best mix: AI does most, a person checks twice

37 — **12 steps:** AI does it all; long run, errors snowball

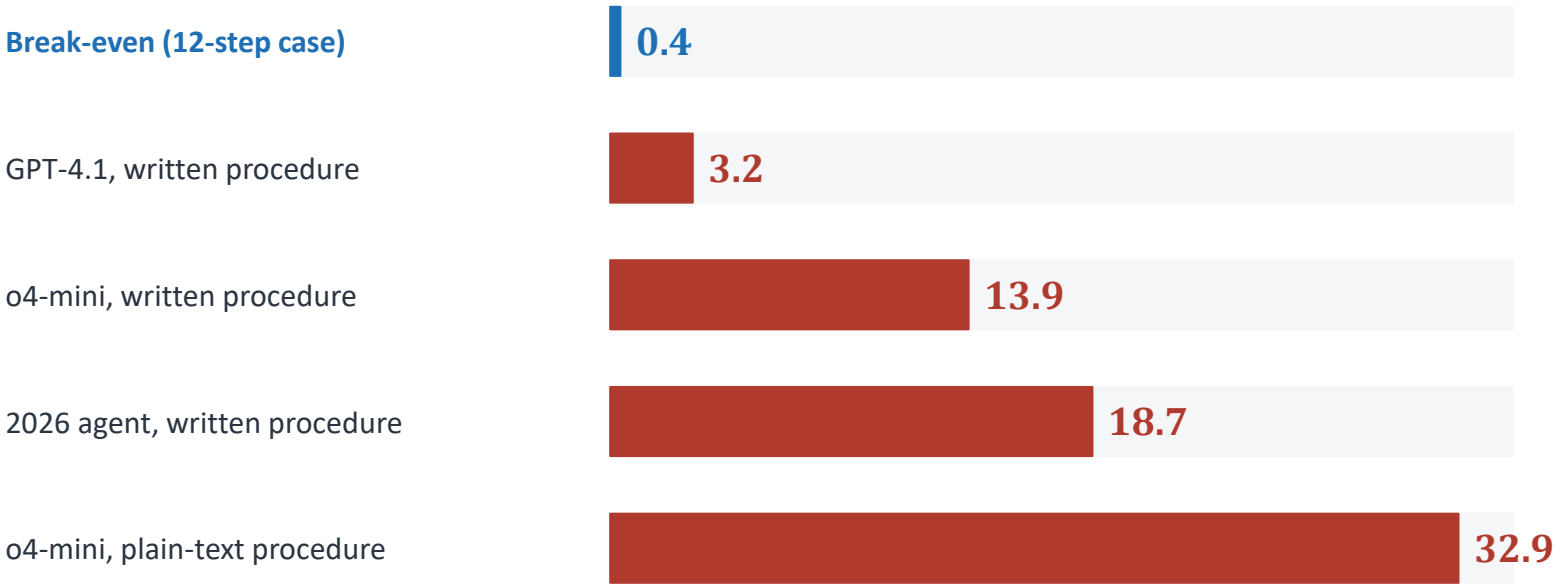
In our data the split was worst in all 4 tests. With our measured hand-off levels, the model shows this dip at every case length tried.

Bringing in AI as an assistant that tells people what to do can leave a firm worse off than before.

How Much Hand-off Error Is Too Much?

The split only pays if hand-off errors stay below a tiny break-even level. Every measured level was far above it.

EXTRA HAND-OFF ERRORS PER 100 STEPS A PERSON CARRIES OUT



Even the best measured level was about 8 times the break-even.

THE BREAK-EVEN GETS SMALLER WHEN

- **The case has more steps**
- **The AI is only a little more reliable than people**
- **Moving steps to the AI saves little labor**

41–79%

cut achieved by writing the procedure down

>98%

cut needed to remove the dip in plain-text tests

Writing the procedure down makes the split less bad. It does not make it safe.

How Long Should an AI Run Before Someone Checks?

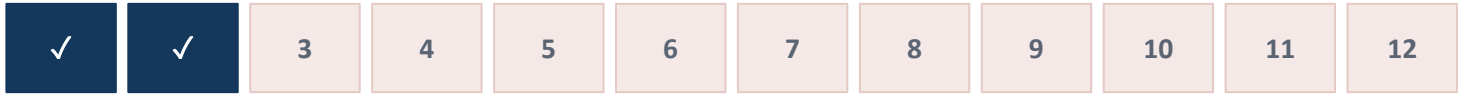
Spread the checks out. And beyond a certain length, do not let even a reliable AI run alone.

RULE 1: SPREAD A FIXED NUMBER OF CHECKS EVENLY

Better: checks spread out, longest unchecked run is 4 steps



Worse: checks bunched at the start, one unchecked run of 10 steps



RULE 2: THE LAST CHECK MATTERS MOST

Removing the last check joins everything into one long unchecked run. Because errors snowball, that long run is far riskier than the two shorter ones it replaces.

So a firm may rightly refuse full autonomy on long cases, even to an AI that is more reliable than a person at every single step.

IN OUR TESTS

4–5 steps

longest run where full autonomy paid, when the checks were customer hand-offs

1–3 steps

when the checks were well-specified review gates

Better checks make supervision more attractive, so full autonomy should stop sooner.

KEEP MEASURING

METR, 2025: the length of task AI agents can finish half the time has doubled about every 7 months since 2019. Safe run lengths will change with each new model, so re-measure them.

Approving full autonomy for short cases does not license it for long ones.

If Reviewers Only Pretend to Check, Risk Rises Fast

A rubber-stamp check costs as much as a real one, but it catches nothing.

WHAT A RUBBER STAMP DOES

Vigilance = the share of checks where the reviewer really looks. A check nobody really looks at joins the stretches on either side into one long unchecked stretch, where errors snowball.

RESEARCH: REVIEWERS DO FOLLOW WRONG AI ADVICE

01 Radiologists, Radiology (2023)

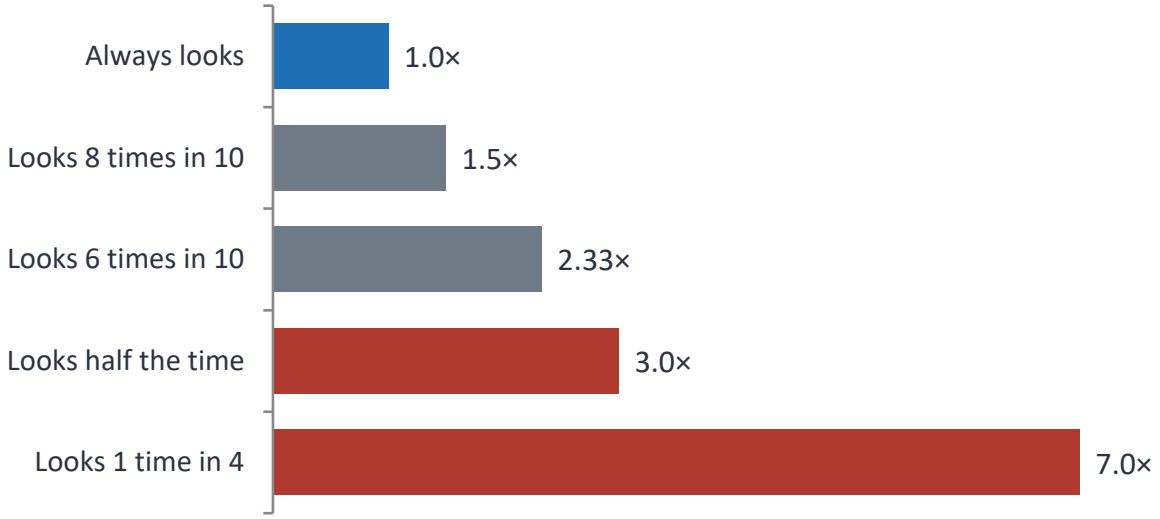
27 radiologists read mammograms with AI advice. When the AI was wrong, accuracy fell from about 80% to under 20% for inexperienced readers, and from 82% to 45.5% for experienced ones.

02 Knowledge workers, CHI 2025

In a survey of 319 workers, more confidence in generative AI went with less critical thinking.

Gartner (Oct 2025): through 2026, loss of critical-thinking skills from GenAI use will push 50% of global organizations to require AI-free skills tests.

HOW MUCH HARDER ERRORS SNOWBALL, BY VIGILANCE



WHAT FIRMS CAN DO

Measure it. Plant known test cases in review packets; count how many are caught.

Pull either lever. Keep reviewers alert, or add tracing and rollback so errors snowball less.

If reviewers look only half the time, errors snowball three times as hard, not twice.

What the Test Looks Like: A Phone Help Desk

The same 114 customer problems, each solved four ways. Only who decides and who acts changes.

ONE CASE, STEP BY STEP (EXAMPLE)

1	**The problem**	A customer says: “My mobile data has stopped working.” The customer is played by an AI.
2	**Hidden causes**	2–9 per case, for example airplane mode is on and the data allowance is used up.
3	**The fix**	1–12 actions. Some only on the phone (turn off airplane mode), some only in the operator’s systems (add data).
4	**Scoring**	A program checks the final state of the phone and the systems. Nobody grades the chat.

SAME CASES, FOUR DESIGNS

AI decides, person does	The AI works out the fix; the customer does the phone steps. *Benchmark: default mode*
AI does it all	The AI has every tool and works alone. *Benchmark: no-user mode*
Plan given, person acts	The AI gets a correct plan and talks the customer through it. *Benchmark: oracle-plan mode*
Plan given, AI acts	The AI carries out a correct plan itself. *Benchmark: oracle plan, no user*

Why a help desk? Gartner (Mar 2025) expects agentic AI to resolve 80% of common customer-service issues without a human by 2029.

4 designs × **2** AI agents × **2** rulebook versions × **114** problems × **4** repeats = **7,296** conversations

Same problem, same customer, same AI. Only the division of work changes, so the design explains the difference.

How We Measure What Gets Lost in the Hand-off

Compare two designs in which the customer does exactly the same steps. Only where the plan comes from differs.

STEP 1: SAME STEPS, TWO SOURCES OF THE PLAN

AI decides, person does
The customer follows the AI's own plan.

vs

Plan given, person acts
The customer follows a correct plan, relayed by the AI.

→

Extra failures per step
= what is lost in the hand-off

HOW WE TELL A WRONG PLAN FROM A LOST HAND-OFF

A wrong plan fails the case once, however many steps it has. A 2-step and a 10-step case suffer the same extra risk.

A lost hand-off adds risk at every step the customer carries out. The extra risk grows with the number of steps.

STEP 2: CHANGE ONLY HOW THE RULES ARE WRITTEN

Plain-text rules
The operator's policy as ordinary prose.

Step-by-step rules
The same policy as a decision procedure.

Who decides and who acts stays the same, so any change in the hand-off loss comes from clearer rules. Results: slide 28.

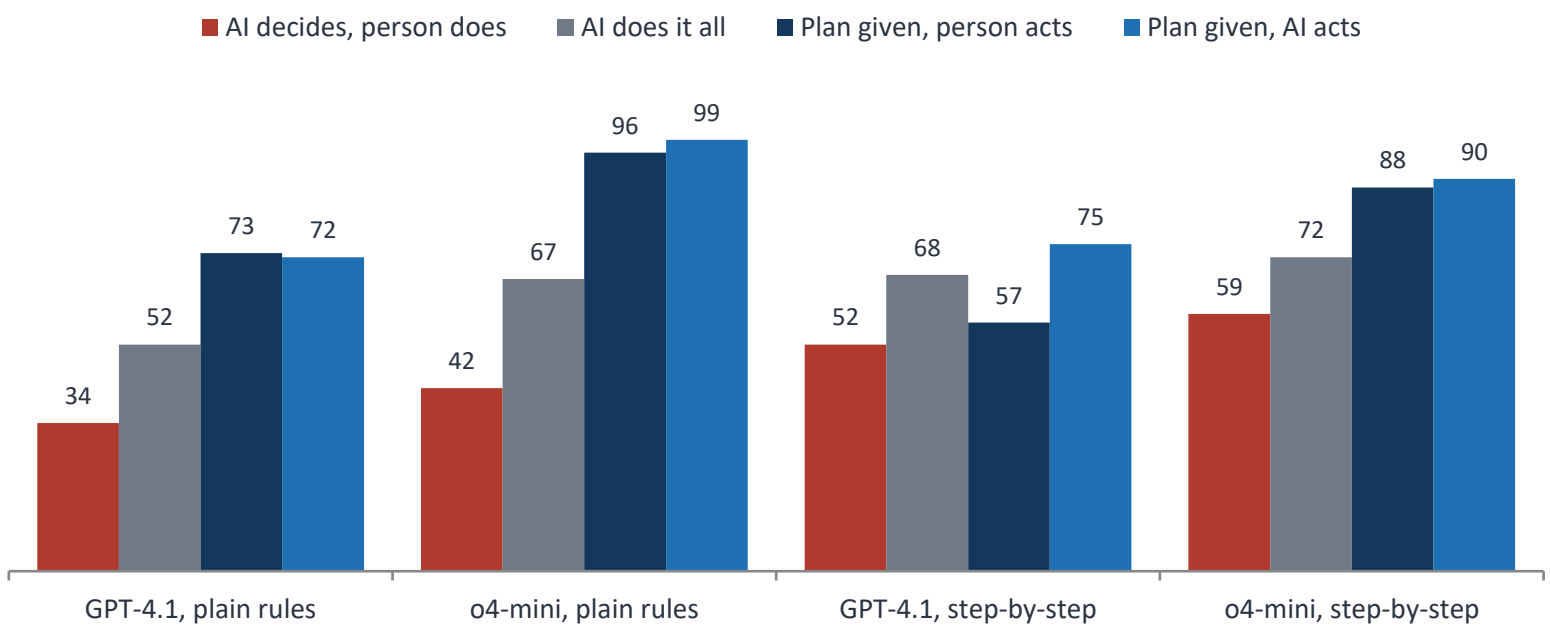
A cautious estimate: even the correct plan passes through the AI's relay. Precision rests on 114 problems, not 7,296 runs.

Same customer, same steps: extra failures per step measure what is lost in the hand-off.

Result 1: Splitting the Work Came Last Every Time

Two AI agents, two rulebooks, the same 114 problems. "AI decides, person does" had the lowest score in all four tests.

PROBLEMS SOLVED OUT OF 100, BY DESIGN



Not just AI agents: a review of 106 experiments found that, on average, person-plus-AI teams did worse than the better of the two alone. The losses came in decision tasks (Nature Human Behaviour, 2024).

IN NUMBERS

4 of 4

tests where "AI decides, person does" came last

23–57 points

behind the best design that kept deciding and doing together

About 2× the cost

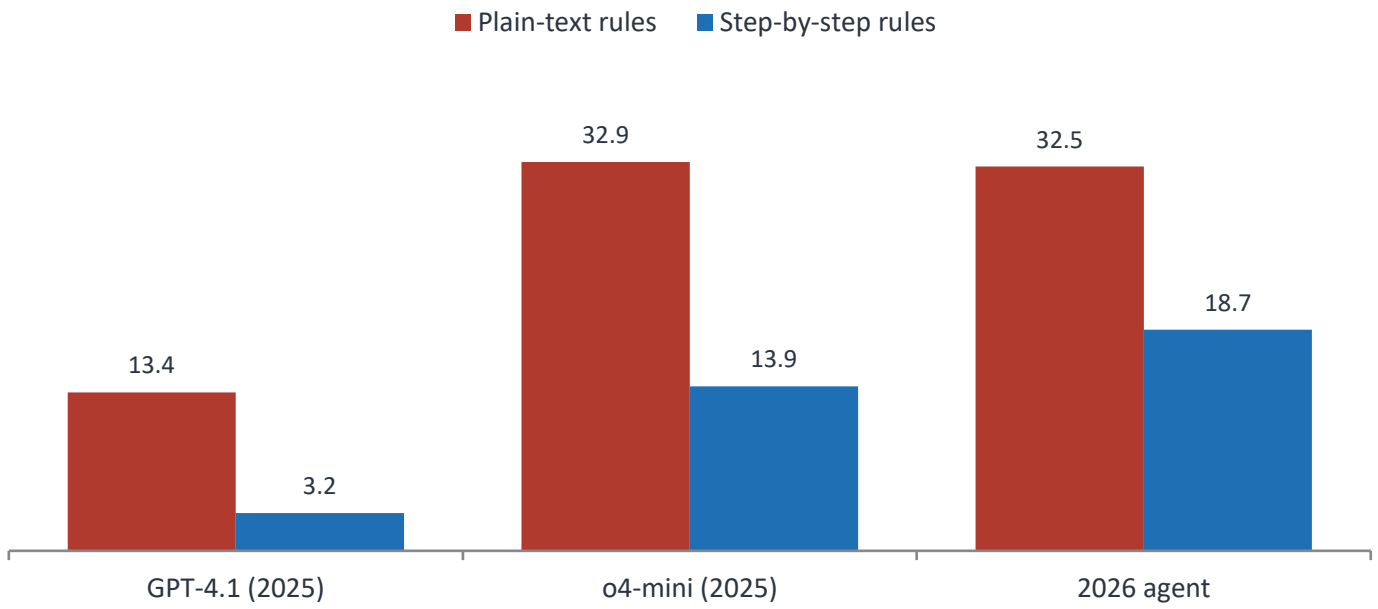
17 vs 8 cents of computing per case, and 2.1 vs 1.4 minutes, against "AI does it all"

Letting the AI also do the steps added 18 points when the AI made the plan, but only 6 when a correct plan was given.

Result 2: Step-by-Step Rules Help, but the Split Still Loses

Clearer rules cut what is lost in the hand-off for every AI tested. The split still came out behind.

HAND-OFF FAILURES PER 100 CUSTOMER STEPS



−76% fewer hand-off failures, GPT-4.1

−58% fewer hand-off failures, o4-mini

−43% fewer hand-off failures, 2026 agent

BUT "AI DECIDES, PERSON DOES" STILL TRAILED

39–54 pts

behind "Plan given, person acts" with plain-text rules

5–29 pts

behind it with step-by-step rules

Not free

Step-by-step rules also lowered success in 3 of 8 design tests, by 8–16 points. Measure the effect in your own workflow.

Other research: on Amazon's SOP-Bench (2025), even with expert-written procedures, the best AI agents' success ranged from 57% to 100% across business domains.

Write the rules step by step before you hand work to AI, but a smaller gap is still a gap.

Question 3: Cheap and Supervised, or Smart and Autonomous?

Economics. A quick show of hands now; we vote again on the last slide.

Can a cheaper AI system under close supervision create more value for a firm than a more advanced AI left to work on its own?

A Smart and autonomous

Better models will soon be reliable enough to run alone.

WHAT RESEARCH SHOWS

METR (2025): the length of task AI agents can finish half the time has doubled about every 7 months since 2019.

B Cheap and supervised

Good design around a modest AI can beat a stronger AI left alone.

WHAT ANALYSTS FORECAST

Gartner forecasts firms will use small, task-specific models 3× more than general LLMs by 2027, and that over 40% of agentic AI projects will be cancelled by end-2027.

WHAT OUR RESULTS ADDS

6 of 6 tests — the same AI did better following a given plan than working alone (+6 to +44 points)

$\sqrt{2}$, not 2× — doubling an AI's reliability edge lengthens its safe solo run only about 1.4 times

The model's answer: sometimes, and predictably. It depends on the hand-off and snowball risks and how long the work runs.

Five New Ideas, and How Sure We Are of Each

What the paper adds to theory, the research each idea builds on, and how strong its evidence is.

	THE IDEA	IN PLAIN WORDS · BUILDS ON	HOW SURE WE ARE
01	**Autonomy is three decisions, not one dial**	Who decides, who does, who checks. Giving the AI the doing was worth far more when it also decided (2025 AIs). *Moves beyond single "levels of automation" (Parasuraman et al., 2000).*	Tested (2025 AIs)
02	**Failures come from how work is divided**	Hand-offs and long unchecked runs fail even with capable people. Each needs its own fix. *Most research blames human limits, e.g. automation bias (Dratsch et al., 2023).*	Tested; snowball partly
03	**When mixing person and AI pays**	A mix beats all-person or all-AI only when errors snowball in long unchecked runs. *Speaks to the automation–augmentation debate (Raisch & Krakowski, 2021; Vaccaro et al., 2024).*	Model; seen for one AI
04	**Reviewer attention has a price**	Keeping reviewers alert and making errors traceable can stand in for each other. *Links to research on complacency with automation (Parasuraman & Manzey, 2010).*	Model only
05	**People who know more can improve the plan**	Involving people may pay when they hold knowledge the AI lacks. *Builds on "specific knowledge" in organizations (Jensen & Meckling, 1992).*	Early hint

Tested: controlled test, designs assigned

Model: follows from the maths, not yet tested on data

Early hint: patterns observed, not proven

The strongest idea is the first: who decides, who does and who checks matters, not only how capable the AI is.

What Three Fields Already Knew, and What AI Changes

Old questions about who should decide what, asked again now that some of the workers are AI.

ORGANIZATION DESIGN

WHAT THE FIELD ALREADY KNEW

Put decisions where the knowledge is. Splitting linked work has a cost. Choose controls by what you can see.

Jensen & Meckling 1992; Thompson 1967; Ouchi 1979. Recent: Puranam 2021 frames human–AI decisions as a design problem.

WHAT AI CHANGES (THIS PAPER)

Puts a price on splitting work between a person and an AI, and treats a human check as a control over a machine.

INFORMATION SYSTEMS

WHAT THE FIELD ALREADY KNEW

People hand work to AI agents and take some back. Autonomy is part of what makes AI hard to manage.

Baird & Maruping, MISQ 2021; Berente et al., MISQ 2021; Raisch & Krakowski, AMR 2021.

WHAT AI CHANGES (THIS PAPER)

Moves from handing over whole tasks to handing over rights within one task, and says when mixing person and AI pays.

ECONOMICS OF DELEGATION

WHAT THE FIELD ALREADY KNEW

What one job is worth depends on the other jobs it comes with. Giving AI authority trades its accuracy against people's effort.

Holmström & Milgrom 1991; Athey, Bryan & Gans 2020.

WHAT AI CHANGES (THIS PAPER)

Same logic, but the link runs through information and errors, not incentives: an AI has no motives to align.

The old questions of organization design return, with a new kind of worker holding some of the decisions.

For Managers: Ask Who Does What, Not How Much Autonomy

Five rules, each tagged by how strong the evidence behind it is.

FROM *"How much autonomy is safe?"* → **TO** ***"Who decides, who does it, and who checks?"***

01	**Don't let the AI decide and a person carry it out**	Keep deciding and doing with one party. The split came last in every test. *Person-plus-AI teams also lose on decision tasks (Nature Human Behaviour, 2024).*	**Best supported**
02	**Write the rules step by step first**	Cut hand-off failures by 43–76%, but lowered success in 3 of 8 tests. Test it first. *Agents still fail with expert-written procedures (SOP-Bench, 2025).*	**Supported**
03	**Limit how long the AI runs unchecked**	In our tests, 4–5 steps; 1–3 with good review gates. Re-check after any redesign. *Agent task length doubles about every 7 months (METR, 2025): re-measure.*	**Partly shown**
04	**Check that reviewers really look**	Plant known test cases. Tracing and rollback can make up for weak review. *Radiologists followed wrong AI advice (Radiology, 2023).*	**Model only**
05	**Keep people who know what the AI does not**	The split result is no reason to remove experts. *Real users did best doing about 30% of the work (slide 30).*	**Caution**

Adding approval steps to an AI's plan adds hand-offs. Where hand-offs are the problem, that makes it worse.

In Practice 1: Let Your Own Data Decide Who Makes the Plan

A simple check, run every review period, on whether the split is still worth keeping.

EVERY REVIEW PERIOD

1 Count the hand-offs

Each time a person carries out a step of the AI's plan, record whether it went wrong.

2 Keep a running score

The share that went wrong, and how sure we are. Few cases means a wide range.

3 Compare with break-even

The error level above which the split loses (slide 19).

THEN DECIDE

Clearly too high, several periods: stop the split; switch to the best other design.

Clearly low: the split is acceptable.

Not sure yet: keep the design and keep counting.

BUILT-IN SAFEGUARDS

Ignores a few bad days

Acts only after enough cases and several periods in a row.

Says if clearer rules can fix it

Reports the cut in errors needed. If it is 98%, rules alone will not do.

Starts again after any change

A new AI model or a new procedure resets the check.

Why your own data, not a vendor's score? On a customer-service benchmark, AI agents' success changed by up to 9 points depending only on which AI played the customer, and simulated customers misjudged results against real people (ACL 2026).

Move the plan only when your own hand-off errors say so, not because of a vendor's benchmark score.

In Practice 2: Check When It Pays, and Check the Checkers

Inside each case: add a check only when it is worth its cost, then test whether reviewers really look.

1 WHEN SHOULD A PERSON LOOK?

WHAT A CHECK SAVES	>	WHAT A CHECK COSTS
The errors it stops from snowballing before the next check		**A person's time, plus one more hand-off if the AI made the plan**

Also check whenever something looks off: a tool error, a change of plan, or a mismatch.

2 IS THE REVIEWER REALLY LOOKING?

Slip a known mistake (a "test case") into some review packets:

Caught	Trust in the review goes up; the AI's run starts fresh
Missed	Trust goes down; the case is flagged
Too many missed	Give reviewers replay logs, or shorten the AI's runs

ALREADY USED ELSEWHERE

Airport security
projects fake threat images onto real bag X-rays and counts how many screeners catch. About 100 such tests per screener give a reliable score (Univ. of Basel, 2023).

WHAT ABOUT AI WATCHING AI?

Gartner (Jun 2025) forecasts that "guardian agents", AI that monitors AI, will take 10–15% of the agentic AI market by 2030. **Caution: in the multi-agent logs of our Study 8, failed verification showed up in over half of the traces.**

Don't count checks. Count the checks that work.

Is Your AI Under Control? Seven Questions for Boards and CIOs

For any workflow where an AI agent can act: the question, a good answer, and a warning sign.

	THE QUESTION	A GOOD ANSWER	WARNING SIGN
1	**Who decides, who does each step, and who checks?**	A written map for every step.	*"The AI recommends; staff carry it out."*
2	**How often do hand-offs from the AI to people fail?**	We log each one and track the rate.	*Nobody records it.*
3	**How long can the AI run before someone looks?**	A limit set from our own error data.	*No limit, or a limit set by feel.*
4	**How do we know reviewers really look?**	We plant test cases and track the catch rate.	*"They sign off on everything."*
5	**Did we write the rules step by step, and did it help?**	Yes, and we compared errors before and after.	*Written, but never measured.*
6	**What makes us re-check the design?**	Any model upgrade, new procedure or rise in errors.	*Nothing, even after an upgrade.*
7	**Who can change who makes the plan, on what evidence?**	The process owner, on our measured errors.	*A vendor's benchmark score.*

Why boards care: Gartner expects 2,000+ legal claims over AI-linked deaths by end-2026, citing weak safeguards. The EU AI Act (Art. 14) requires high-risk AI to allow effective human oversight, including awareness of over-reliance on its output.

Governing AI agents starts with a map of who decides, does and checks, not a list of tools.

Before We Discuss: Answer Again on Your Phone

Three minutes. Then we compare the room's answers before and after the talk.

ON THE SAME PAGE AS BEFORE, OPEN "YOUR VIEW"

1. **Access ladder**
   For each workflow you played: from 0 No access to 5 Act alone.
2. **Unchecked steps**
   How many steps may an agent run before a person checks?
3. **The freedom question again**
   How much freedom should AI agents get today?
4. **One rule**
   A rule you would write for your organization, and work you would never hand over.

Link: https://who-does-the-work.streamlit.app/

Your answers are today's data

Discussion: Our Three Questions, Revisited

Vote again. Has the evidence changed your answer?

QUESTION 1 · STRATEGY

Where will advantage come from: better agents or better design?

WHAT WE SAW

Same AI, same tasks: the split scored 23–57 points below the best design that kept deciding and doing together.

Leans to B, better design
Seen in controlled tests

QUESTION 2 · OB

Fewer managers, or a different kind of oversight?

WHAT WE SAW

New oversight work appears: placing checks, measuring hand-offs, testing reviewers.

Leans to B, different oversight
From the model; not yet tested on people

QUESTION 3 · ECONOMICS

Cheap and supervised, or smart and autonomous?

WHAT WE SAW

The same AI did better following a given plan than working alone in 6 of 6 tests.

Sometimes B: it depends on the risks and run length
Plan was correct: a best case

Don’t ask “how much autonomy?” Ask “who decides, who does, who checks, and on what evidence?”

Inside the Data: Two Real Problems, One Repeat

Real records from the τ²-bench telecom logs used in the paper. The same problem is run under every design.

PROBLEM A · 2 HIDDEN FAULTS

"My mobile data is not working, or it is very slow."

Hidden faults: mobile data switched off; monthly data allowance used up.

Correct plan (2 actions): customer turns mobile data on (phone) → agent adds 2 GB of data (operator's system).

Solved when: data is on, the speed test shows "excellent" and 2 GB was added.

PROBLEM B · 4 HIDDEN FAULTS

"My phone has been showing 'No Service' for the past few hours."

Hidden faults: airplane mode on; network (APN) settings broken; line suspended for an overdue bill; SIM card loose.

Correct plan (8 actions): airplane mode off → reset APN → reboot → agent sends bill → customer pays → agent resumes line → reboot → reseat SIM.

Solved when: the phone shows a signal and the bill is paid.

REPEAT 1 · GPT-4.1 · PLAIN-TEXT RULES

DESIGN	PROBLEM A	PROBLEM B
AI decides, person does	✓ Solved	✗ Failed
AI does it all	✓ Solved	✓ Solved
Plan given, person acts	✓ Solved	✓ Solved
Plan given, AI acts	✓ Solved	✓ Solved

What went wrong in the split: the AI cleared the bill, airplane mode and the SIM, but never asked the customer to reset the network settings. Still "No Signal", so it handed the case to a human.

AI agent: GPT-4.1 (or o4-mini)

Customer: GPT-4.1 following a script

Rulebook: Plain-text (or step-by-step)

Repeat: 1 of 4; same random seed in every design

4 designs × **2** AI agents × **2** rulebooks × **114** problems × **4** repeats = **7,296** conversations

Same problem, same customer, same repeat: only who decided and who acted changed, and only the split missed a fault.

Thank You

Questions and comments are welcome.

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